In [15]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

In [7]:
```
df=pd.read_csv("Reliance.csv")
```

In [9]:
```
df
```

Out[9]:

	Date	Open	High	Low	Close	Adj Close	Vo
0	2015-11-18	463.799988	465.649994	454.975006	456.000000	436.671021	5142
1	2015-11-19	459.450012	469.350006	458.625000	467.375000	447.563873	5569
2	2015-11-20	467.000000	476.399994	462.774994	473.424988	453.357422	5167
3	2015-11-23	475.000000	478.950012	473.100006	476.875000	456.661224	4800
4	2015-11-24	476.500000	485.799988	475.524994	483.850006	463.340515	6768
...	...	...	...	...	...	...	
1228	2020-11-10	2077.000000	2090.000000	2041.199951	2084.550049	2084.550049	17045
1229	2020-11-11	2089.000000	2095.000000	1978.099976	1997.199951	1997.199951	26178
1230	2020-11-12	1981.000000	2008.449951	1965.000000	1980.000000	1980.000000	18481
1231	2020-11-13	1982.000000	2036.650024	1981.750000	1996.400024	1996.400024	20946
1232	2020-11-17	2085.000000	2085.000000	1985.000000	1993.250000	1993.250000	21479

1233 rows × 7 columns

In [11]:
```
df.ffill
```

Out[11]:
```
<bound method NDFrame.ffill of             Date         Open         Hig
h          Low        Close  \
0     2015-11-18   463.799988   465.649994   454.975006   456.000000
1     2015-11-19   459.450012   469.350006   458.625000   467.375000
2     2015-11-20   467.000000   476.399994   462.774994   473.424988
3     2015-11-23   475.000000   478.950012   473.100006   476.875000
4     2015-11-24   476.500000   485.799988   475.524994   483.850006
...          ...          ...          ...          ...          ...
1228  2020-11-10  2077.000000  2090.000000  2041.199951  2084.550049
1229  2020-11-11  2089.000000  2095.000000  1978.099976  1997.199951
1230  2020-11-12  1981.000000  2008.449951  1965.000000  1980.000000
1231  2020-11-13  1982.000000  2036.650024  1981.750000  1996.400024
1232  2020-11-17  2085.000000  2085.000000  1985.000000  1993.250000

        Adj Close      Volume
0      436.671021   5142766.0
1      447.563873   5569752.0
2      453.357422   5167930.0
3      456.661224   4800026.0
4      463.340515   6768886.0
...           ...         ...
1228  2084.550049  17045147.0
1229  1997.199951  26178477.0
1230  1980.000000  18481466.0
1231  1996.400024  20946864.0
1232  1993.250000  21479385.0

[1233 rows x 7 columns]>
```

In [13]:
```
df.bfill
```

Out[13]:
```
<bound method NDFrame.bfill of             Date         Open         Hig
h          Low        Close  \
0     2015-11-18   463.799988   465.649994   454.975006   456.000000
1     2015-11-19   459.450012   469.350006   458.625000   467.375000
2     2015-11-20   467.000000   476.399994   462.774994   473.424988
3     2015-11-23   475.000000   478.950012   473.100006   476.875000
4     2015-11-24   476.500000   485.799988   475.524994   483.850006
...          ...          ...          ...          ...          ...
1228  2020-11-10  2077.000000  2090.000000  2041.199951  2084.550049
1229  2020-11-11  2089.000000  2095.000000  1978.099976  1997.199951
1230  2020-11-12  1981.000000  2008.449951  1965.000000  1980.000000
1231  2020-11-13  1982.000000  2036.650024  1981.750000  1996.400024
1232  2020-11-17  2085.000000  2085.000000  1985.000000  1993.250000

        Adj Close      Volume
0      436.671021   5142766.0
1      447.563873   5569752.0
2      453.357422   5167930.0
3      456.661224   4800026.0
4      463.340515   6768886.0
...           ...         ...
1228  2084.550049  17045147.0
1229  1997.199951  26178477.0
1230  1980.000000  18481466.0
1231  1996.400024  20946864.0
1232  1993.250000  21479385.0

[1233 rows x 7 columns]>
```

In [17]:
```
df.isnull()
```

Out[17]:

	Date	Open	High	Low	Close	Adj Close	Volume
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
1228	False	False	False	False	False	False	False
1229	False	False	False	False	False	False	False
1230	False	False	False	False	False	False	False
1231	False	False	False	False	False	False	False
1232	False	False	False	False	False	False	False

1233 rows × 7 columns

In [19]:
```
df.isnull().sum()
```

Out[19]:
```
Date         0
Open         1
High         1
Low          1
Close        1
Adj Close    1
Volume       1
dtype: int64
```

In [21]:
```
df.describe()
```

Out[21]:

	Open	High	Low	Close	Adj Close	Volum
count	1232.000000	1232.000000	1232.000000	1232.000000	1232.000000	1.232000e+0
mean	1036.035414	1048.479285	1023.015573	1035.164502	1024.471417	9.981743e+0
std	463.719976	469.906022	455.946191	462.301697	466.231636	7.603435e+0
min	459.000000	465.649994	444.049988	453.274994	434.061554	1.283997e+0
25%	551.925003	561.393768	546.000000	553.049988	540.720703	5.555522e+0
50%	959.099976	972.474976	946.675018	960.949982	946.843903	7.560520e+0
75%	1309.024963	1321.649963	1292.274963	1308.337494	1298.972321	1.140781e+0
max	2325.000000	2369.350098	2310.550049	2324.550049	2324.550049	6.523004e+0

In [23]:
```
df.dropna()
```

Out[23]:

	Date	Open	High	Low	Close	Adj Close	V
0	2015-11-18	463.799988	465.649994	454.975006	456.000000	436.671021	5142
1	2015-11-19	459.450012	469.350006	458.625000	467.375000	447.563873	5569
2	2015-11-20	467.000000	476.399994	462.774994	473.424988	453.357422	5167
3	2015-11-23	475.000000	478.950012	473.100006	476.875000	456.661224	4800
4	2015-11-24	476.500000	485.799988	475.524994	483.850006	463.340515	6768
...	...	...	...	...	...	...	
1228	2020-11-10	2077.000000	2090.000000	2041.199951	2084.550049	2084.550049	17045
1229	2020-11-11	2089.000000	2095.000000	1978.099976	1997.199951	1997.199951	26178
1230	2020-11-12	1981.000000	2008.449951	1965.000000	1980.000000	1980.000000	18481
1231	2020-11-13	1982.000000	2036.650024	1981.750000	1996.400024	1996.400024	20946
1232	2020-11-17	2085.000000	2085.000000	1985.000000	1993.250000	1993.250000	21479

1232 rows × 7 columns

In [41]:
```
df['Date']=pd.to_datetime(df['Date'])
```

```
Date          datetime64[ns]
Open                 float64
High                 float64
Low                  float64
Close                float64
Adj Close            float64
Volume               float64
dtype: object
```

In [55]:
```
print(df[['Close']])
```

```
            Close
0      456.000000
1      467.375000
2      473.424988
3      476.875000
4      483.850006
...           ...
1228  2084.550049
1229  1997.199951
1230  1980.000000
1231  1996.400024
1232  1993.250000

[1233 rows x 1 columns]
```

In [45]:
```
df
```

Out[45]:

	Date	Open	High	Low	Close	Adj Close	Vo
0	2015-11-18	463.799988	465.649994	454.975006	456.000000	436.671021	5142
1	2015-11-19	459.450012	469.350006	458.625000	467.375000	447.563873	5569
2	2015-11-20	467.000000	476.399994	462.774994	473.424988	453.357422	5167
3	2015-11-23	475.000000	478.950012	473.100006	476.875000	456.661224	4800
4	2015-11-24	476.500000	485.799988	475.524994	483.850006	463.340515	6768
...	...	...	...	...	...	...	
1228	2020-11-10	2077.000000	2090.000000	2041.199951	2084.550049	2084.550049	17045
1229	2020-11-11	2089.000000	2095.000000	1978.099976	1997.199951	1997.199951	26178
1230	2020-11-12	1981.000000	2008.449951	1965.000000	1980.000000	1980.000000	18481
1231	2020-11-13	1982.000000	2036.650024	1981.750000	1996.400024	1996.400024	20946
1232	2020-11-17	2085.000000	2085.000000	1985.000000	1993.250000	1993.250000	21479

1233 rows × 7 columns

In [43]:
```
print(df.dtypes)
```

```
Date          datetime64[ns]
Open                 float64
High                 float64
Low                  float64
Close                float64
Adj Close            float64
Volume               float64
dtype: object
```

In [57]:
```
df[['Date','Close']]
```

Out[57]:

	Date	Close
0	2015-11-18	456.000000
1	2015-11-19	467.375000
2	2015-11-20	473.424988
3	2015-11-23	476.875000
4	2015-11-24	483.850006
...	...	...
1228	2020-11-10	2084.550049
1229	2020-11-11	1997.199951
1230	2020-11-12	1980.000000
1231	2020-11-13	1996.400024
1232	2020-11-17	1993.250000

1233 rows × 2 columns

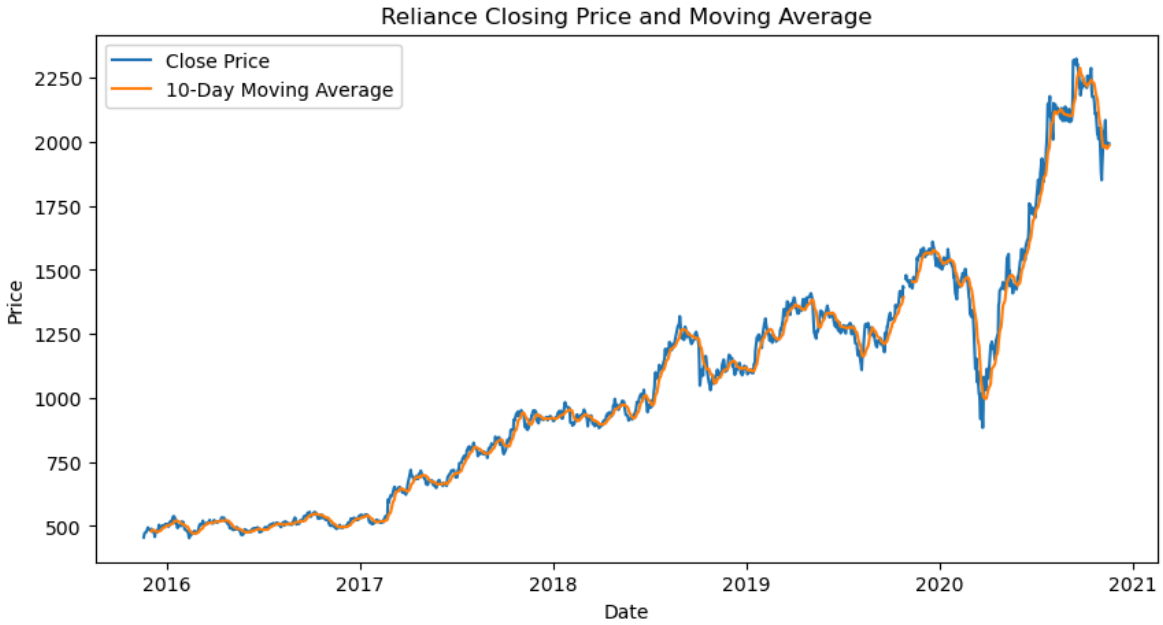
In [61]:

```
plt.plot(df['Date'], df['Close'])
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.show()
```

In [63]:

```
df['MA10'] = df['Close'].rolling(window=10).mean()
plt.figure(figsize=(10,5))
plt.plot(df['Date'], df['Close'], label='Close Price')
plt.plot(df['Date'], df['MA10'], label='10-Day Moving Average')
plt.xlabel("Date")
plt.ylabel("Price")
```

```
plt.title("Reliance Closing Price and Moving Average")
plt.legend()
plt.show()
```



In [65]:
```
df.plot(x='Date',y='Close')
```

Out[65]: <Axes: xlabel='Date'>

In [93]:
```
df['moving variable1']=df['Close'].rolling(window=50).mean()
```

In [95]:
```
print(df[['Close','moving variable1']].tail())
```

```
            Close  moving variable1
1228  2084.550049       2157.511006
1229  1997.199951       2155.710005
1230  1980.000000       2152.746006
1231  1996.400024       2150.432004
1232  1993.250000       2148.752004
```

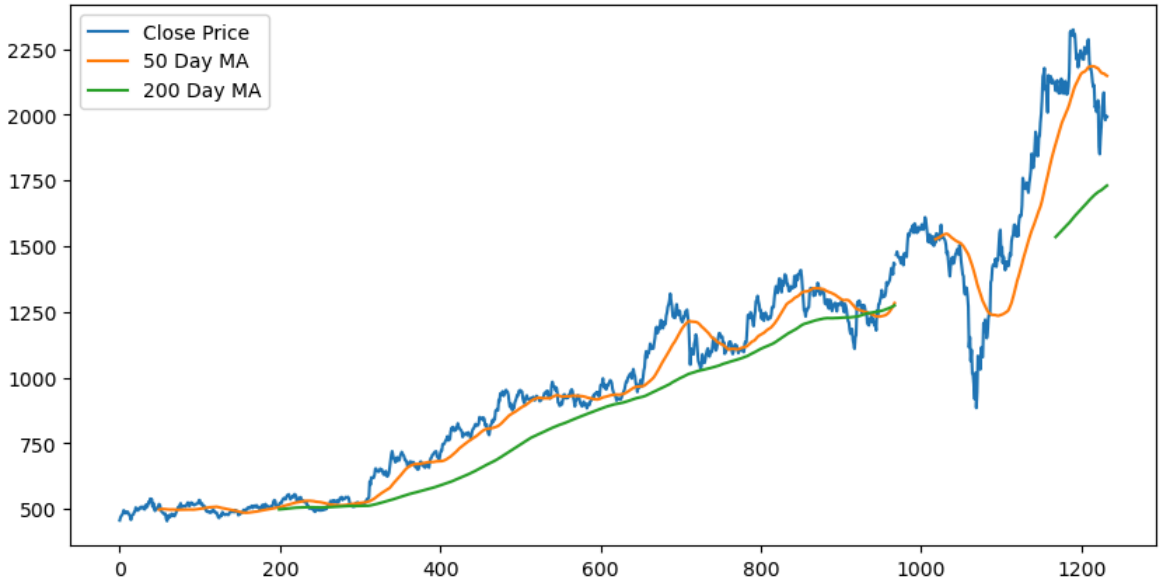
In [97]:
```
df['moving variable2']=df['Close'].rolling(window=200).mean()
```

In [99]:
```
print(df[['Close','moving variable2']].tail())
```

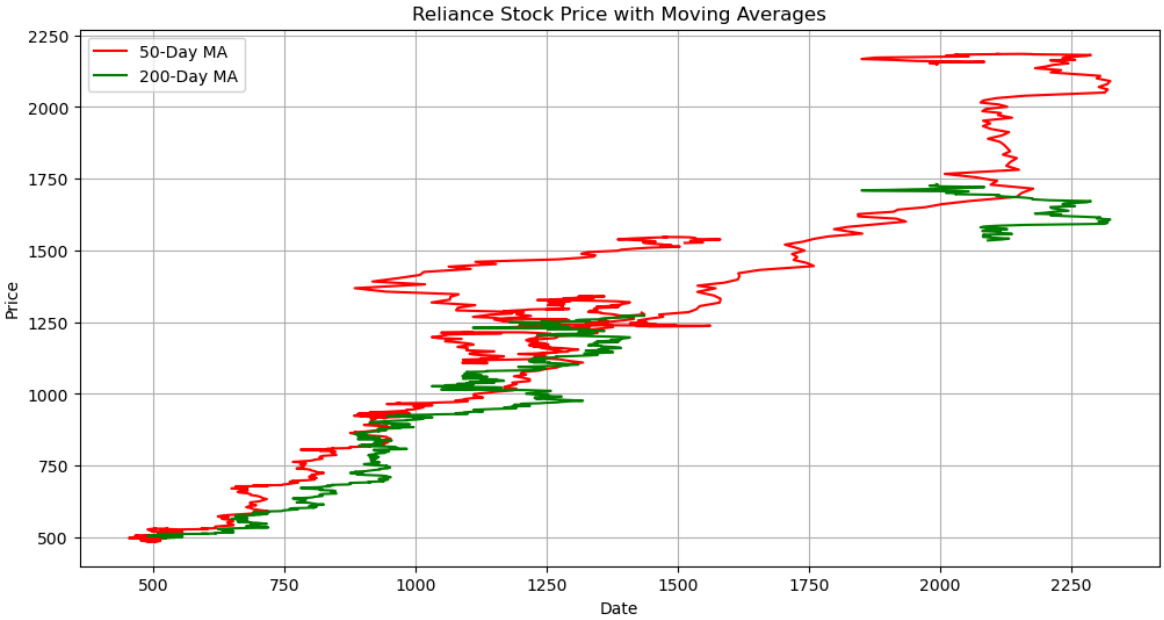
```
            Close  moving variable2
1228  2084.550049       1721.059958
1229  1997.199951       1723.411708
1230  1980.000000       1725.703957
1231  1996.400024       1728.153207
1232  1993.250000       1730.760707
```

In [101…
```
plt.figure(figsize=(10,5))
plt.plot(df['Close'], label='Close Price')
plt.plot(df['moving variable1'], label='50 Day MA')
plt.plot(df['moving variable2'], label='200 Day MA')

plt.legend()
plt.show()
```



In [116…
```
plt.figure(figsize=(12,6))
plt.plot(df['Close'], df['moving variable1'], label='50-Day MA', color='r
plt.plot(df['Close'], df['moving variable2'], label='200-Day MA', color='
plt.title("Reliance Stock Price with Moving Averages")
plt.xlabel("Date")
plt.ylabel("Price")
plt.legend()
plt.grid(True)
plt.show()
```



In [104…
```
df.rename(columns={'Date': 'ds'}, inplace=True)
```

In [108…
```
df.rename(columns={'moving variable':'y'}, inplace=True)
```

In [118…
```
df
```

Out[118…

	ds	Open	High	Low	Close	Adj Close	V
0	2015-11-18	463.799988	465.649994	454.975006	456.000000	436.671021	5142
1	2015-11-19	459.450012	469.350006	458.625000	467.375000	447.563873	5569
2	2015-11-20	467.000000	476.399994	462.774994	473.424988	453.357422	5167
3	2015-11-23	475.000000	478.950012	473.100006	476.875000	456.661224	4800
4	2015-11-24	476.500000	485.799988	475.524994	483.850006	463.340515	6768
...	...	...	...	...	...	...	
1228	2020-11-10	2077.000000	2090.000000	2041.199951	2084.550049	2084.550049	17045
1229	2020-11-11	2089.000000	2095.000000	1978.099976	1997.199951	1997.199951	26178
1230	2020-11-12	1981.000000	2008.449951	1965.000000	1980.000000	1980.000000	18481
1231	2020-11-13	1982.000000	2036.650024	1981.750000	1996.400024	1996.400024	20946
1232	2020-11-17	2085.000000	2085.000000	1985.000000	1993.250000	1993.250000	21479

1233 rows × 11 columns

In [156…
```
!pip install prophet
```

```
4646.71s - pydevd: Sending message related to process being replaced timed
-out after 5 seconds
Requirement already satisfied: prophet in ./anaconda3/lib/python3.12/site-
packages (1.3.0)
Requirement already satisfied: cmdstanpy>=1.0.4 in ./anaconda3/lib/python
3.12/site-packages (from prophet) (1.3.0)
Requirement already satisfied: numpy>=1.15.4 in ./anaconda3/lib/python3.1
2/site-packages (from prophet) (2.2.6)
Requirement already satisfied: matplotlib>=2.0.0 in ./anaconda3/lib/python
3.12/site-packages (from prophet) (3.9.2)
Requirement already satisfied: pandas>=1.0.4 in ./anaconda3/lib/python3.1
2/site-packages (from prophet) (2.2.3)
Requirement already satisfied: holidays<1,>=0.25 in ./anaconda3/lib/python
3.12/site-packages (from prophet) (0.92)
Requirement already satisfied: tqdm>=4.36.1 in ./anaconda3/lib/python3.12/
site-packages (from prophet) (4.66.5)
Requirement already satisfied: importlib_resources in ./anaconda3/lib/pyth
on3.12/site-packages (from prophet) (6.5.2)
Requirement already satisfied: stanio<2.0.0,>=0.4.0 in ./anaconda3/lib/pyt
hon3.12/site-packages (from cmdstanpy>=1.0.4->prophet) (0.5.1)
Requirement already satisfied: python-dateutil<3,>=2.9.0.post0 in ./anacon
da3/lib/python3.12/site-packages (from holidays<1,>=0.25->prophet) (2.9.0.
post0)
Requirement already satisfied: contourpy>=1.0.1 in ./anaconda3/lib/python
3.12/site-packages (from matplotlib>=2.0.0->prophet) (1.3.1)
Requirement already satisfied: cycler>=0.10 in ./anaconda3/lib/python3.12/
site-packages (from matplotlib>=2.0.0->prophet) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in ./anaconda3/lib/python
3.12/site-packages (from matplotlib>=2.0.0->prophet) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in ./anaconda3/lib/python
3.12/site-packages (from matplotlib>=2.0.0->prophet) (1.4.4)
Requirement already satisfied: packaging>=20.0 in ./anaconda3/lib/python3.
12/site-packages (from matplotlib>=2.0.0->prophet) (24.2)
Requirement already satisfied: pillow>=8 in ./anaconda3/lib/python3.12/sit
e-packages (from matplotlib>=2.0.0->prophet) (11.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in ./anaconda3/lib/python
3.12/site-packages (from matplotlib>=2.0.0->prophet) (3.2.0)
Requirement already satisfied: pytz>=2020.1 in ./anaconda3/lib/python3.12/
site-packages (from pandas>=1.0.4->prophet) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in ./anaconda3/lib/python3.1
2/site-packages (from pandas>=1.0.4->prophet) (2023.3)
Requirement already satisfied: six>=1.5 in ./anaconda3/lib/python3.12/site
-packages (from python-dateutil<3,>=2.9.0.post0->holidays<1,>=0.25->prophe
t) (1.16.0)
```

In [193…

```python
from prophet import Prophet
```

In [195…

```python
model=Prophet()
model.fit(df)
```

```
10:44:48 - cmdstanpy - INFO - Chain [1] start processing
10:44:48 - cmdstanpy - INFO - Chain [1] done processing
```

Out[195…

```
<prophet.forecaster.Prophet at 0x7fb632afd700>
```

In [197…

```python
future = model.make_future_dataframe(periods=180)
```

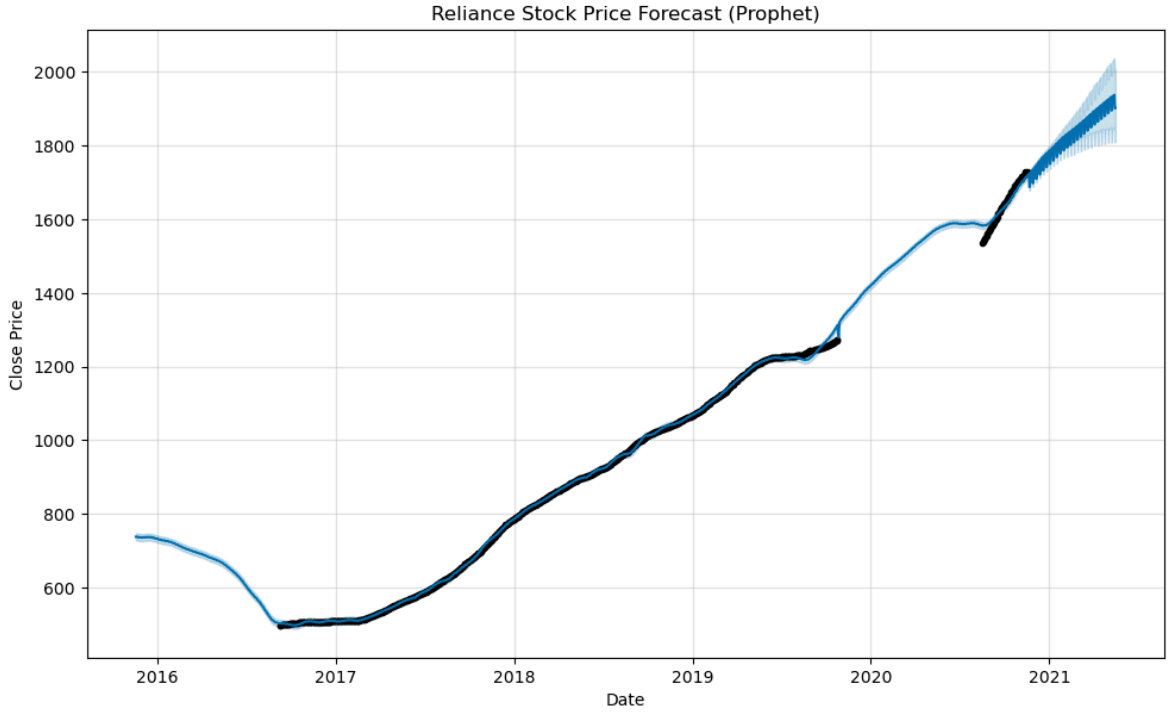
In [199…

```python
forecast = model.predict(future)
```

In [201…

```
import matplotlib.pyplot as plt

fig = model.plot(forecast)
plt.title("Reliance Stock Price Forecast (Prophet)")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.show()
```



In [ ]:

In [ ]:

In [ ]: